WONG JIA XI   B032310495

BITS 3S1G2

INTERNET TECHNOLOGY

```
Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-5TVEQFQP
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Radmin VPN:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Famatech Radmin VPN Ethernet Adapter
   Physical Address. . . . . . . . . : 02-50-82-A5-68-FE
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : fdfd::1a55:9543(Preferred)
   Link-local IPv6 Address . . . . . : fe80::ded0:de23:6770:ba11%8(Preferred)
   IPv4 Address. . . . . . . . . . . : 26.85.149.67(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.0.0.0
   Default Gateway . . . . . . . . . : 26.0.0.1
   DHCPv6 IAID . . . . . . . . . . . : 889344130
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-68-48-B4-30-F6-EF-77-7B-
DA
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 30-F6-EF-77-7B-DB
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

```
Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
#2
   Physical Address. . . . . . . . . : 32-F6-EF-77-7B-DA
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
   Physical Address. . . . . . . . . : 30-F6-EF-77-7B-DA
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.1.104(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Friday, March 27, 2026 9:43:42 PM
   Lease Expires . . . . . . . . . . : Saturday, March 28, 2026 9:43:42 PM
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DNS Servers . . . . . . . . . . . : 1.1.1.1
                                       1.0.0.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

```
C:\Users\User>ping www.utem.edu.my

Pinging www.utem.edu.my [172.67.72.33] with 32 bytes of data:
Reply from 172.67.72.33: bytes=32 time=87ms TTL=59
Reply from 172.67.72.33: bytes=32 time=99ms TTL=59
Reply from 172.67.72.33: bytes=32 time=86ms TTL=59
Reply from 172.67.72.33: bytes=32 time=102ms TTL=59

Ping statistics for 172.67.72.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 86ms, Maximum = 102ms, Average = 93ms
```

```
C:\Users\User>msg * /server:192.168.1.105
'msg' is not recognized as an internal or external command,
operable program or batch file.
```

Msg command is not available for my Windows edition. This is common on Windows 11 Home, the msg command is only available on Windows Pro/Enterprise.

Notes

Self Review Question 1

1. Hostname

Display computer name on the network.

2. Ipconfig.

Show IP address, Subnet mask, and gateway for all adapters.

Computer's IP address	IPv4 = 192.168.1.104 IPv6 : fdfd :: 1a55 : 9543
Computer Name	LAPTOP - 5TVE8F8P
Router's IP Address	192.168.1.1
IP setting	Dynamic DHCP Enabled.
MAC number	30-F6-EF-77-7B-PA
Public IP Address	180.75.252.199
Check with Friend's IP address. Are they similar? Why? :	My friend IP address 180.75.233.134 is similar to my laptop IP address. They share same prefix 180.75.X.X which mean they are same ISP.

3. ping — Test reachability of a host by sending ICMP packets and measuring round trip time.

request time out — reachable but not responding.

destination host unreachable — packet couldn't reach destination.

Notes

4. pathping

Combine ping and tracert.

5. msg

Send message to other user on same network / LAN.

6. Traceroute

Show every router hop from your PC to destination

7 hops to flink website.

PC (192.168.1.104)

↓

192.168.1.1

↓

100.99.127.254

↓

10.233.65.54

↓

* * *

↓

13335.sgw.equinix.com (27.111.228.132)

↓

172.69.117.55.

↓

104.26.14.166

Notes

7. arp — Displays / modifies the ARP table

8. nbtstat — Show NetBIOS over TCP/IP statistics and name tables.

9. netstat — Displays active TCP connections, listening ports, and networking statistics.

10. nslookup — Queries DNS servers to resolve hostnames to IP addresses.

11. route — Displays / modifies IP routing table.

12. netview — List computers and shared resources visible on local network.

Notes Self Review Question 2.

1. Why we need port number?
- A PC can run many task at same time.
- Port numbers allows OS to direct incoming network traffic to correct application.

2. Common Port Num

Application	Port Num
FTP	21
HTTP	80
HTTPs	443
Email – SMTP	25
Email – POP3	110
Telnet	23
SSH	22
DNS	53